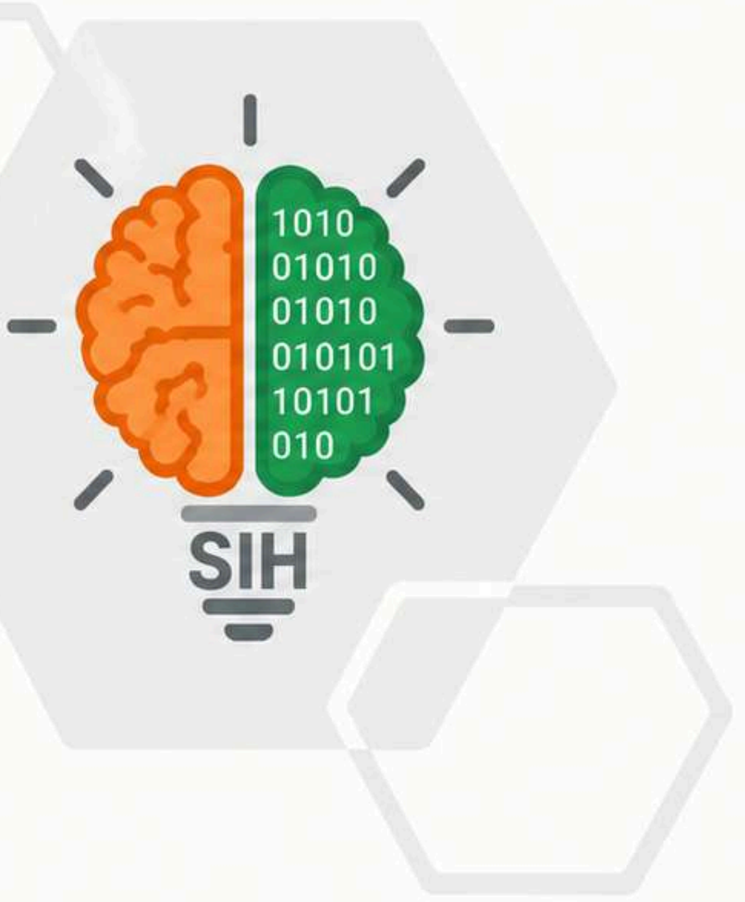
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Nirikshan (निरीक्षण)

- Problem Statement ID - 26102
- Problem Statement Title - Development of an AI-powered system to detect anomalies, fraud, and inefficiencies in MPLAD Schemes
- Theme - Smart Automation
- PS Category - Software
- Team Name - Lateratus Solutions



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PROPOSAL: AI-Based MPLADS Monitoring Platform.

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1 Data Ingestion & Validation (Pydantic)

Fetches real-time data from **multiple** sources and validates it using Pydantic for accuracy and consistency.

Govt APIs · Transport Feeds · Open Data → Pydantic Validation

2 Analytics & Prediction

Uses ML models to detect risks, predict congestion and optimize resource allocation.

Real-time Tracking → ML Models (Fault & Risk Detection) → Cost & Resource Benchmarking

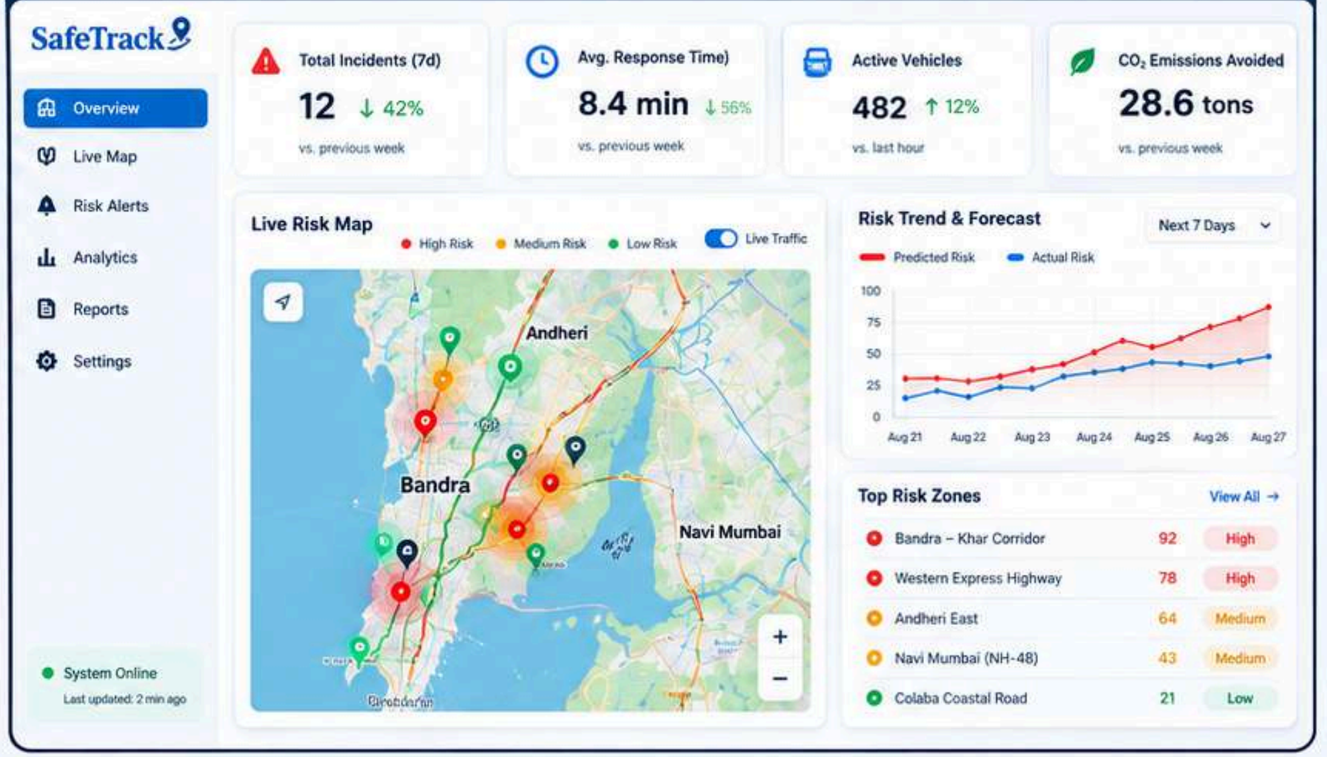
3 Risk Synthesis & Alerts

Combines insights, generates risk scores and provides actionable alerts via a smart dashboard.

Risk Scoring → Smart Alerts → Web + Mobile Dashboard

- Faster Response Times
- Reduced Accidents
- Lower Emissions
- Safer Communities

INTEGRATED DASHBOARD VIEW (PROTOTYPE)



- **EXPLAINABLE AI (XAI):** Clear audit justifications.
- **CPU-OPTIMIZED ARCHITECTURE:** Standard NIC Cloud efficiency.
- **DUPLICATE PREVENTION:** via Perceptual Hashing.

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TECHNICAL APPROACH

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CORE TECHNOLOGY STACK

Languages
Python 3.11 | TypeScript | SQL (PostgreSQL)

Backend & Ingestion
FastAPI | Pydantic | Celery + Redis

Data & Analytics
scikit-learn | pandas | numpy | SHAP Engine

Vision & Forensics
ImageHash (pHash) | Pillow / ExifRead | YOLOv8n (Nano)

Database & Cache
PostgreSQL (JSONB) | Redis Hot Cache

THE PARALLEL DETECTION PIPELINE (Defense-in-Depth)

Public Portal / CPWD DSR

01 Ingestion & Gatekeeping

- **Data Ingestion**
  Public Portal / CPWD DSR Data Pipeline
- **Data Cleaning**
  Pydantic Schema Validation (Blocks malformed entries)

02 Core Analytics Tracks (Parallel Execution)

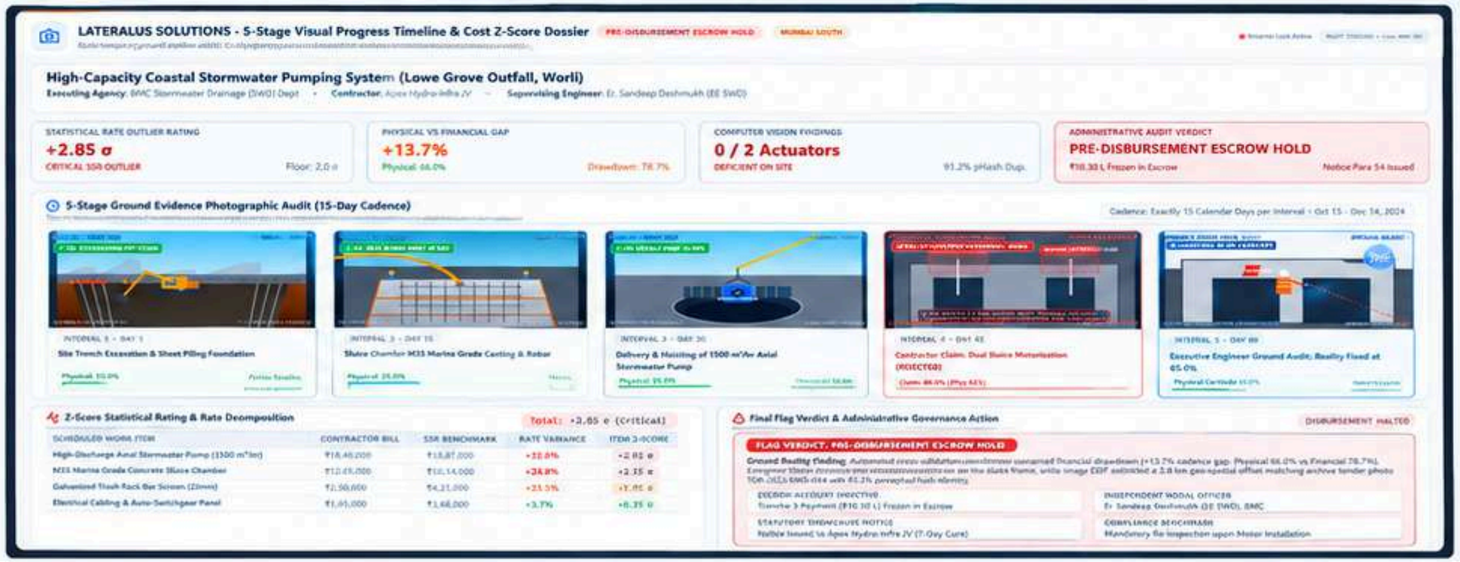
- **Module 1: Utilization**
  - Isolation Forest
  - Fund-Dumping Check
  - Spending Velocity
- **Module 2: Cost Overrun**
  - Z-Score Outlier
  - CPWD/DSR Benchmark
  - Rate Manipulation
- **Module 3: Image/Photo**
  - Perceptual Hashing
  - EXIF Metadata Audit
  - YOLOv8n Asset Count

03 Risk Synthesis & Human Triage Interface

- **Weighted Aggregation**
  Risk Score = 0.25(Util) + 0.40(Cost) + 0.35(Photo)
- **Explainable AI Engine**
  SHAP Engine generating human-readable audit justifications
- **Target Delivery Desk**
  Dynamic Dashboard sorting the prioritized Risk-Ranked Review Queue

Audit Teams & Decision Makers

WORKING PROTOTYPE (DASHBOARD)



PROTOTYPE vs. PRODUCTION INFRASTRUCTURE

Compute / Host	Frontend	Orchestration	Security Layer
Local Docker Compose / Railway	Vercel Preview Deployment	Docker Containers	Env-Based Secrets
NIC Cloud (MeghRaj)	eSAKSHI Subdomain (Govt SSL)	Kubernetes Horizontal Scaling	NIC-Issued Auth + VAPT Compliance

Development (Local) → Staging (Vercel) → Production (NIC Cloud)

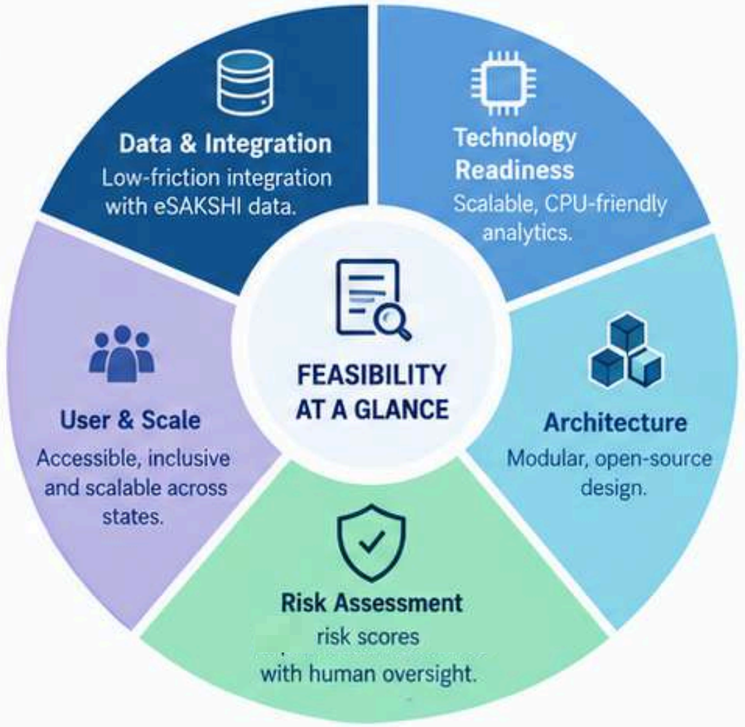
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FEASIBILITY AND VIABILITY

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01 MULTI-DIMENSIONAL FEASIBILITY ANALYSIS

Evaluating the solution from every angle.



THE RESULT

A practical, scalable and reliable solution built for real-world impact. →

02 CRITICAL ARCHITECTURE CHALLENGES

Complex problems. Targeted solutions.

LIMITED GROUND TRUTH
Scarcity of labeled fraud data limits supervised model accuracy.
Leads to model drift and false negatives.

FRAGMENTED DATA
Data silos prevent comprehensive cross-scheme validation.
Incomplete view of citizen transactions.

GRIEVANCE MANIPULATION
Grievance volume is not a proxy for genuine irregularity.
Risk of misclassification and noise.

LOCAL COST VARIATION
Highly variable local costs make benchmarks unreliable.
Skews detection and comparison across regions.

SATELLITE LIMITATIONS
Environmental factors (clouds) constrain satellite verification.
Incomplete or delayed imagery.

03 INSTITUTIONAL MITIGATION

Stronger systems. Higher trust.

ISOLATION FOREST
Unsupervised anomaly detection for pattern flagging.
Detects unusual patterns early.

PROGRESSIVE DATA INTEGRATION
Adaptive data confidence scoring based on source quality.
Improves accuracy with multi-source data.

GRIEVANCE VERIFICATION
Corroboration required for all citizen-reported anomalies.
Reduces noise and false positives.

CONTEXT-AWARE ANALYSIS
Regional and historical benchmarking for cost accuracy.
Accounts for local and temporal variations.

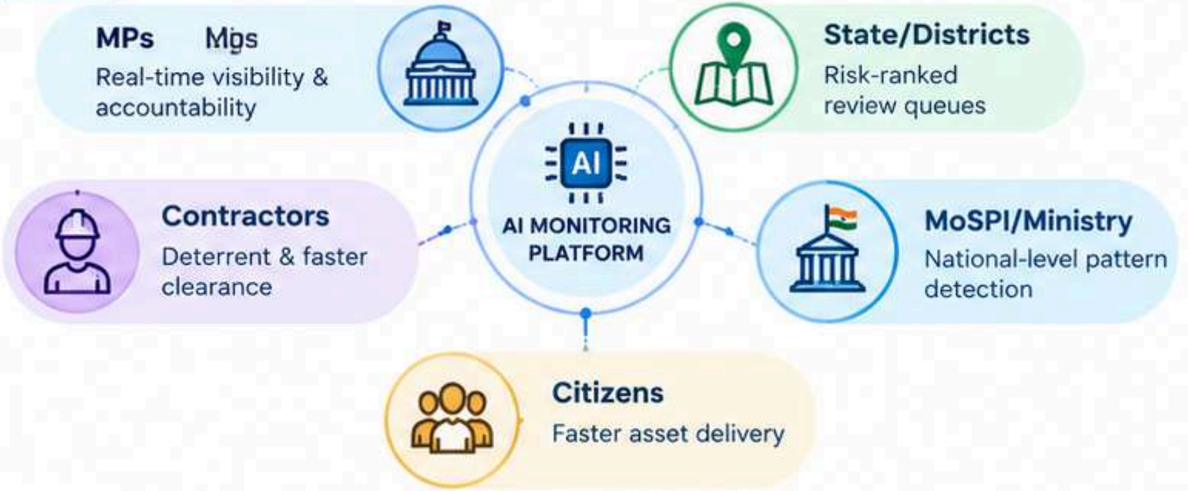
MULTI-SOURCE VERIFICATION
Triangulate satellite data with ground-level evidence.
Completes the picture with evidence.

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IMPACT AND BENEFITS

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Audience Impact



Solution Benefits

Social	✓ Trust restored
Economic	✓ Leakage prevented
Environmental	✓ Waste avoided
Governance	✓ Explainable & scalable

AI vs Manual Monitoring: A Paradigm Shift

Parameter	Current Manual Monitoring	Proposed AI Platform	AI ENGINE / METHODS
Timing	✗ Post-facto →	✓ Real-time	Streaming / real-time
Scope	✗ 15–20% →	✓ 100% automated	CPWD/DSR automation
Rate	✗ Manual →	✓ Automated	pHash + EXIF + YOLO
Verification	✗ Subjective →	✓ Explainable verification	Isolation Forest / perceptual hashing
Fraud	✗ Duplication →	✓ Fraud detection	SHAP + localized benchmarks
Cross-Scheme	✗ Weak →	✓ Cross-scheme deduplication	Similarity matching
Explainability	✗ Grievance-driven →	✓ Role-based dashboards	SHAP
Cost	✗ Decision-heavy / manual →	✓ Low-cost CPU stack	Efficient CPU inference

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RESEARCH AND REFERENCES

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1. Government & Domain Sources 🏛️

- MPLADS Scheme Guidelines — Ministry of Statistics & Programme Implementation (MoSPI) — *mplads.gov.in*
- eSAKSHI Portal (fund-flow & work-tracking system, live since April 2023) — *mplads.mospi.gov.in*
- CPWD Delhi Schedule of Rates (DSR) 2023 — *cpwd.gov.in/Documents/cpwd_publication.aspx*
- Open Government Data (OGD) Platform — *data.gov.in* (for public dataset sourcing/benchmarking)

2. Secondary & Academic Research Sources 📖

- NITI Aayog — Reports on Infrastructure Development & Local Area Impact Evaluation — *niti.gov.in*
- Comptroller and Auditor General (CAG) of India — Performance Audit Reports on MPLADS Implementation — *cag.gov.in*